The conjectures in *A Bayesian Algorithm for Optimal Dating Expenditure*: proofs, counterexamples and corrected statements

Companion note

30 October 2026

Abstract

The proposal states five conjectures (Conjectures 1–5) and leaves one question open: whether the value of information can be negative once refusals sting (Section 4.4). None of the five conjectures holds as stated in the core model. Each is false in general, which is shown by an explicit counterexample, and each has a true core, which is proved under conditions stated exactly. The open question is answered: in the core model, information about compatibility always has non-negative value. So does information about willingness that comes from monotone signals, such as the guide's cues. A non-monotone signal about willingness can have strictly negative value under all three models of the sting. Every number in this note is reproduced by `verify.py`, which also runs randomised checks of each theorem.

Summary

Claim	Verdict	What is proved
Conjecture 1 (threshold rules)	False in general (Prop. 8)	Threshold rules hold under a single condition (M) (Theorem 6). The condition is automatic when the after-date choice does not depend on what the date reveals. With two types it reads $v_j(C_H - C_L) \geq \tilde{\kappa}\,\lvert\psi(C_H) - \psi(C_L)\rvert$ (Corollary 7).
Conjecture 2 (outside option)	True for Phase C; false on the day in general	Raising U shrinks the set of beliefs at which the agent asks for another date. On the day, raising U makes continuing *less* attractive at beliefs where the agent would cut contact after the date, and *more* attractive where they would ask for another date (Lemma 10, Theorem 11).
Conjecture 3 (relationship value)	First part: true for Phase C, false on the day in general. Second part: false	Raising R has the mirror-image effect of raising U (Theorem 13). The investment value of a step peaks where the after-date choice is closest to indifference, not where beliefs are least settled. It vanishes as beliefs settle (Proposition 15).
Conjecture 4 (tight budgets)	Premise rarely holds; conclusion partly true	Concavity gives the shadow-value claim by definition, but entry prices make V non-concave in B. Without any concavity, thresholds rise as the budget falls in the core model. Under concavity, spend falls with the budget. "Cheaper opening activities" is false (Proposition 16).
Conjecture 5 (spending to learn)	First and last parts true; "peaks under uncertainty" false	Optimal spend is never below the Proposition 1 level, under all three sting models. It converges to that level as beliefs settle. The premium need not be largest when beliefs are least settled (Proposition 21).
Open question, Sec. 4.4	Answered	EVSI ≥ 0 for every signal about C, and for every MLRP signal about q, under D1, D2 and KR. A non-MLRP signal about q can have negative EVSI under D1, D2 and KR (Theorem 24, Examples 25 and 26).

1 Setting

Notation is that of the proposal. Throughout, b denotes the agent's belief, $\mathbb{E}_b$ expectation under it, $p = \varphi_j \mathbb{E}_b[q]$ the probability of a yes at step j, and

$$\hat{p}(p) \;=\; p - \kappa_m(p), \qquad m \in \{\mathrm{D1}, \mathrm{D2}, \mathrm{KR}\},$$

the *effective continuation weight*: $\kappa_{\mathrm{D1}} = \eta(1-p)$, $\kappa_{\mathrm{D2}} = \eta p(1-p)$, $\kappa_{\mathrm{KR}} = \eta(\Lambda - 1)p(1-p)$.

Standing assumption S1 (Core model)**.** Assumptions A1–A12 hold, including independent priors (A5). The value after the date is truncated after one more date, as in (19)–(21), with $h(c, q_D) = \bar{h}\, c\, q_D$.

Standing assumption S2 (Ratings)**.** After step j the agent observes a rating y whose likelihood $P_j(y \mid c)$ has the monotone likelihood ratio property (MLRP) in c, and updates by Bayes' rule. The leading case is $y \sim \mathrm{Bin}(m_j, C)$ with integer m_j. Beta beliefs then stay Beta: $\mathrm{Beta}(\alpha, \beta) \mapsto \mathrm{Beta}(\alpha + y, \beta + m_j - y)$.

Remark 1 (The fractional likelihood)*.* The fractional Binomial (2) is a tempered quasi-likelihood. It does not by itself define a distribution for the rating, so expected posterior moments need not equal prior moments. Under it, $\mathbb{E}[\mathbb{E}_{b'} f(C) \mid b] = \mathbb{E}_b f(C)$ holds for $f(c) = c$ when $\mathbb{E}[\tilde{y} \mid C] = m_a C$, but not for nonlinear f. The results below that use the martingale property for nonlinear functions of C therefore hold exactly with integer m_a and exact Bayes, and approximately with fractional m_a. The results that use only monotonicity carry over unchanged.

Standing assumption S3 (No sting without a stake)**.** The expected feeling of an answer is $\kappa_m(p)\,\Delta^+$ with $\Delta^+ = \max\{\Delta, 0\}$. The proposal assumes $\Delta > 0$ throughout; this extends it to $\Delta \le 0$ without turning a refusal into a gain.

Standing assumption S4 (Willingness to meet again)**.** Where stated, q_D is known, so $P_D = q_D$. This is part of the hypothesis of Conjecture 1. It is used elsewhere only to make the after-date value linear-plus-option in the belief about C (Lemma 2); the same results hold if q_D is uncertain and $\eta = 0$ in Phase C.

Write $\tilde{\kappa} = (\hat{p}(q_D))^+ = (q_D - \kappa_m(q_D))^+$, and define

$$\begin{aligned} a(c) &= h(c)\, c\, R + \big(1 - h(c)\big) U, \\ \psi(c) &= \big(1 - h(c)\big)\big[\delta_D\big(h(c)\, c\, R + (1 - h(c))U\big) - U\big], \qquad h(c) = \bar{h}\, q_D\, c, \end{aligned}$$

and the *Phase-C index* $\Psi(b) = \mathbb{E}_b[\psi(C)]$.

Lemma 2 (After-date value)**.** *Under S1, S3 and S4,*

$$V_{\mathrm{end}}(b) \;=\; \mathbb{E}_b[a(C)] \;+\; \tilde{\kappa}\; (\Psi(b))^+ .$$

After a date that did not become a relationship, the agent asks for another date if and only if $\tilde{\kappa} > 0$ *and* $\Psi(b) > 0$*.*

Proof. With q_D known, a yes to another date says nothing about C, so in (20) $b_{\mathrm{yes}} = b$ and the stake is $\Delta_D = V_D(b_{\mathrm{not}}) - U$. If $\Delta_D \ge 0$, asking is worth $U + \hat{p}(q_D)\Delta_D$, so $V_C(b_{\mathrm{not}}) = U + \tilde{\kappa}\Delta_D$. If $\Delta_D < 0$, asking is worth $U + q_D\Delta_D < U$, so $V_C = U$. Hence $V_C(b_{\mathrm{not}}) = U + \tilde{\kappa}\,\Delta_D^+$. The event "no relationship" has likelihood $1 - h(C)$, so $\mathbb{E}_{b_{\mathrm{not}}}[f(C)]\,(1 - \mathbb{E}_b h) = \mathbb{E}_b[(1 - h(C))f(C)]$ for every f. Applying this to (21) gives $(1 - \mathbb{E}_b h)(V_D(b_{\mathrm{not}}) - U) = \mathbb{E}_b\psi = \Psi(b)$. Since $1 - \mathbb{E}_b h > 0$, $(1 - \mathbb{E}_b h)\Delta_D^+ = \Psi(b)^+$. Substituting into (19):

$$V_{\mathrm{end}}(b) = \mathbb{E}_b[hCR] + (1 - \mathbb{E}_b h)U + \tilde{\kappa}\,\Psi(b)^+ = \mathbb{E}_b[a] + \tilde{\kappa}\,\Psi(b)^+. \qquad \square$$

So V_{end} is linear in b plus a convex option term. The option has a kink exactly where the after-date choice switches between cutting contact and asking for another date.

Lemma 3 (A refusal carries no news; effective continuation weight)**.** *Under S1 and S3, the loss from a no is $L = 0$, so $\ell = 0$ and $\Delta = G$, at every step of every date. Writing*

$$G_j(b, B) = \sup_{s \in S_j(B)} \Big\{ W_j(b; s) - c_j^{\text{time}} + \mathbb{E}\big[V_{j+1}(b', B - \sigma(k_j + s)) \mid b_{\text{yes}}\big] \Big\} - V_{\text{end}}(b)$$

(with $G_j = -\infty$ if $S_j(B) = \emptyset$), the Bellman equation (8) is equivalent to

$$V_j(b, B) = V_{\text{end}}(b) + \hat{p}_j^+ \, G_j(b, B)^+, \qquad \hat{p}_j = \hat{p}(\varphi_j \mathbb{E}_b q). \tag{1}$$

The agent asks if and only if $\hat{p}_j > 0$ and $G_j \geq 0$, and the optimal venue maximises the term in braces.

Proof. By Lemma 2, or directly from (19)–(21), V_{end} depends on b only through its marginal on (C, q_D). By A4 and A5 an answer is informative only about q, which is independent of (C, q_D). Hence $V_{\text{end}}(b_{\text{no}}) = V_{\text{end}}(b_{\text{yes}}) = V_{\text{end}}(b)$, so $L = 0$. At spend s, let $G(s)$ be the term in braces minus $V_{\text{end}}(b)$. Asking is worth $V_{\text{end}}(b) + pG(s) - \kappa(p)G(s)^+$. This is $V_{\text{end}} + \hat{p}\, G(s)$ when $G(s) \geq 0$, and is below V_{end} when $G(s) < 0$. Maximising over s and comparing with ending gives (1). □

Since $\hat{p}^+ \in [0, 1]$, (1) can be rewritten as

$$V_j = \max\Big\{ V_{\text{end}}(b),\ \hat{p}_j^+ \big(G_j + V_{\text{end}}(b)\big) + (1 - \hat{p}_j^+) V_{\text{end}}(b) \Big\}.$$

This is the Bellman equation of an *optimal stopping problem*. After each ask the date continues with probability $\hat{p}^+$ and pays the step's flow. With probability $1 - \hat{p}^+$ it ends at the current belief about C, and when the date ends the agent receives V_{end} at the terminal belief. Call this fictitious measure $\tilde{\mathbb{P}}$, with expectation $\tilde{\mathbb{E}}$. Because $\hat{p}$ does not depend on C, and ratings are Bayesian, the belief about C is a $\tilde{\mathbb{P}}$-martingale. So for every policy π and stopping time $\tau \leq J + 1$,

$$\tilde{\mathbb{E}}^{\pi}\big[\mathbb{E}_{b_\tau} f(C)\big] = \mathbb{E}_b f(C) \qquad \text{for every bounded } f. \tag{2}$$

This is the tool behind most of what follows.

Remark 4 (Closed form of Proposition 1). With $\bar{s}_j = k_j$ and $x = \mathbb{E}[C] v_j$, the net value of a step at the optimal spend is

$$W_j = x \ln \frac{x}{\sigma \lambda k_j} \quad \text{if } x \geq \sigma \lambda k_j, \qquad W_j = x - \sigma \lambda k_j \quad \text{otherwise.}$$

Discretionary spend is therefore positive exactly when the step is worth taking at its entry price.

2 Conjecture 1: threshold rules

Conjecture 1 concerns the one-dimensional case, with q and q_D known. Then $p_j = \varphi_j q$ and $\hat{p}_j$ are constants, and the agent asks at step j iff $\hat{p}_j > 0$ and $G_j \geq 0$ (Lemma 3). At each step the belief lies in a totally ordered family F_j:

- two types, $b = \pi = P(C = C_H)$; or
- Beta, $b = \text{Beta}(\alpha, n_j - \alpha)$, with $n_j = n_1 + \sum_{i<j} m_i$ fixed by the step.

Both families are ordered by the likelihood ratio, and $\mathbb{E}_b C$ is strictly increasing along them. With MLRP ratings:

(i) the posterior $T_j(b, y)$ is nondecreasing in b and y;
(ii) the predictive distribution of y is increasing in b in the first-order stochastic sense;
(iii) hence $b \mapsto \mathbb{E}[\phi(T_j(b, y)) \mid b]$ is nondecreasing for every nondecreasing ϕ on F_{j+1}.

Definition 5. The *investment value* of step j is

$$I_j(b) = \mathbb{E}[V_{\text{end}}(T_j(b,y)) \mid b] - V_{\text{end}}(b) = \tilde{\kappa}\Big(\mathbb{E}\big[\Psi(T_j(b,y))^+ \mid b\big] - \Psi(b)^+\Big) \geq 0.$$

The second expression follows from Lemma 2 and the martingale property; the sign follows from convexity.

Theorem 6 (Threshold rules)**.** *Assume S1–S4 with q known, and assume for every j:*

$$\textbf{(M)} \qquad I_j(b_2) - I_j(b_1) \;\geq\; -v_j\big(\mathbb{E}_{b_2}C - \mathbb{E}_{b_1}C\big) \qquad \text{for all } b_1 \leq b_2 \text{ in } F_j.$$

Then for every step j and budget B, $G_j(\cdot, B)$ is nondecreasing on F_j. The set of beliefs at which the agent asks is therefore $\{b \in F_j : \mathbb{E}_b C \geq c_j^*(B)\}$, a threshold on $\mathbb{E}[C]$, possibly at 0 or above 1.

Proof. Let $O_j = V_j - V_{\text{end}} = \hat{p}_j^+ G_j^+$, and note $O_{J+1} \equiv 0$. Suppose $O_{j+1}(\cdot, B')$ is nondecreasing on F_{j+1} for every B'; this holds for $j = J$. Since V_{end} does not depend on B,

$$G_j(b,B) = \sup_{s \in S_j(B)} \Big\{ W_j(b;s) - c_j^{\text{time}} + \mathbb{E}\big[O_{j+1}(T_j(b,y), B - \sigma(k_j + s)) \mid b\big] \Big\} + I_j(b).$$

Fix $b_1 \leq b_2$ and $\varepsilon > 0$, and let s_1 be ε-optimal at b_1. Then:

- $W_j(b_2; s_1) - W_j(b_1; s_1) = v_j\big(1 + g(s_1)\big)(\mathbb{E}_{b_2}C - \mathbb{E}_{b_1}C) \geq v_j(\mathbb{E}_{b_2}C - \mathbb{E}_{b_1}C)$, because $g \geq 0$;
- the O_{j+1} term is nondecreasing in b, by (iii) and the induction hypothesis;
- (M) bounds the fall in I_j.

Hence $G_j(b_2, B) - G_j(b_1, B) \geq -\varepsilon$ for every $\varepsilon > 0$. So $G_j(\cdot, B)$ is nondecreasing, and then so is $O_j(\cdot, B) = \hat{p}_j^+ G_j^+$. Induction completes the proof. □

Condition (M) asks the consumption value of the step to rise with $\mathbb{E}[C]$ at least as fast as the step's investment value can fall. The investment value is the value of what the step reveals for the after-date choice. The next corollary gives conditions on primitives.

Corollary 7. *Condition (M), and so the conclusion of Theorem 6, holds in each of the following cases.*

(a) The after-date choice does not depend on what the date reveals: $\tilde{\kappa} = 0$, *or* ψ *has constant sign on the support of beliefs. Then* $I_j \equiv 0$.

(b) Two types: $v_j(C_H - C_L) \geq \tilde{\kappa}\,|\psi(C_H) - \psi(C_L)|$ *for every* j. *This does not depend on the rating technology.*

(c) Beta beliefs: Ψ *is nondecreasing on every* F_j, *and* $\tilde{\kappa} L_j \leq v_j$, *where*

$$L_j = \sup_{b_1 < b_2 \in F_j} \frac{\Psi(b_2) - \Psi(b_1)}{\mathbb{E}_{b_2}C - \mathbb{E}_{b_1}C}.$$

If ψ *is nondecreasing on* $[0,1]$, *both parts hold whenever* $\tilde{\kappa} \sup_c \psi'(c) \leq \min_j v_j$.

Proof. (a) If $\psi \geq 0$ on the support, $\Psi \geq 0$ at every posterior, so $I_j = \tilde{\kappa}(\mathbb{E}[\Psi(b')] - \Psi(b)) = 0$ by (2). The case $\psi \leq 0$ is the same, and $\tilde{\kappa} = 0$ is immediate.

(b) Write $x_0 = \psi(C_L)$, $x_1 = \psi(C_H)$ and $P_y^H = P_j(y \mid C_H)$, $P_y^L = P_j(y \mid C_L)$. Then

$$I_j(\pi)/\tilde{\kappa} = \sum_y \big(\pi P_y^H x_1 + (1-\pi) P_y^L x_0\big)^+ - \big(x_0 + \pi(x_1 - x_0)\big)^+.$$

This is continuous and piecewise linear in π. Where it is differentiable, its slope is

$$\sum_{y \in A(\pi)} \big(P_y^H x_1 - P_y^L x_0\big) - \mathbf{1}\{\Psi(\pi) > 0\}(x_1 - x_0),$$

where $A(\pi)$ is the set of y whose term is positive.

- If $x_0 < 0 < x_1$, every summand is positive, so the slope is at least $-(x_1 - x_0)$.
- If $x_1 < 0 < x_0$, every summand is negative, so the sum is at least $\sum_y (P_y^H x_1 - P_y^L x_0) = x_1 - x_0$, and the indicator term is non-negative.
- If x_0 and x_1 have the same sign, (a) applies.

So the slope of I_j in π is at least $-\tilde{\kappa}|x_1 - x_0|$. Since $\mathbb{E}_\pi C = C_L + \pi(C_H - C_L)$, (M) holds under the stated inequality.

(c) For $b_1 \le b_2$ in F_j, write

$$\frac{I_j(b_2) - I_j(b_1)}{\tilde{\kappa}} = \Big[\mathbb{E}\big(\Psi^+(b') \mid b_2\big) - \mathbb{E}\big(\Psi^+(b') \mid b_1\big)\Big] - \Big[\Psi^+(b_2) - \Psi^+(b_1)\Big].$$

The first bracket is non-negative by (iii), because Ψ^+ is nondecreasing on F_{j+1}. The second is at most $(\Psi(b_2) - \Psi(b_1))^+ \le L_j(\mathbb{E}_{b_2} C - \mathbb{E}_{b_1} C)$. If ψ is nondecreasing, first-order dominance gives

$$\Psi(b_2) - \Psi(b_1) = \int \psi'(c)\,\big(F_{b_1}(c) - F_{b_2}(c)\big)\,dc \le \sup \psi' \cdot \big(\mathbb{E}_{b_2} C - \mathbb{E}_{b_1} C\big),$$

and Ψ is nondecreasing. □

Proposition 8 (Conjecture 1 fails without (M))**.** *Take two types $C \in \{0.2, 0.8\}$ and one step left, with $v = 50$, $k = 30$, $c^{\text{time}} = 0$, $\sigma = \lambda = 1$ and one Bernoulli rating ($m = 1$). Let $q = 0.9$, D1 with $\eta = 0.2$, $q_D = 0.8$, $\bar{h} = 0.5$, $\delta_D = 0.9$, $R = 6000$, $U = 1000$. Then the agent asks at the last step on $(0.071, 0.232) \cup (0.667, 1]$ and ends on the rest of $[0, 1]$. The set of asking beliefs is not an interval, and no threshold rule is optimal.*

Proof. The primitives give:

- $\psi(0.2) = -78.75$ and $\psi(0.8) = 676.19$, so $\Psi(\pi) = 0$ at $\pi_k = 0.1043$;
- $\tilde{\kappa} = 0.8 - 0.2 \cdot 0.2 = 0.76$ and $\hat{p} = 0.9 - 0.2 \cdot 0.1 > 0$;
- $s^* = \max(50\,\mathbb{E}C - 30, 0)$, which is 0 for $\mathbb{E}C \le 0.6$, so $W(\pi) = 50\,\mathbb{E}_\pi C - 30$ there.

Consider four regions.

- For $\pi \ge 0.318$, the posterior after a low rating, $\pi/(\pi + 4(1 - \pi))$, is above π_k. So Ψ^+ is linear on the support of the posteriors, and $I(\pi) = 0$. Then $G = W < 0$ for $\mathbb{E}_\pi C < 0.6$, that is, for $\pi < 0.667$; and $G \ge 0$ above.
- At $\pi = \pi_k$, only a high rating leads to asking for another date. So

  $$I(\pi_k) = \tilde{\kappa}\big(\pi_k \cdot 0.8 \cdot 676.19 + (1 - \pi_k) \cdot 0.2 \cdot (-78.75)\big) = 32.17,$$

  while $W(\pi_k) = -16.87$, giving $G(\pi_k) = 15.29 > 0$.
- At $\pi = 0.05$, $G = -9.32 < 0$.

G therefore changes sign three times. `verify.py` computes G on a fine grid and confirms the switch points. □

The mechanism is the after-date option. Near π_k the agent is close to indifferent between cutting contact and asking for a second date. The last step's rating settles that choice, so the step is worth taking for what it reveals, even though it is not worth its price as an evening out. At somewhat higher beliefs the second date is already decided, the information is worthless, and the agent ends. Here (M) fails badly: $v(C_H - C_L) = 30$, while $\tilde{\kappa}|\psi(C_H) - \psi(C_L)| = 574$.

Remark 9. When (M) fails, the asking set at each step is still a finite union of intervals in $\mathbb{E}[C]$. A card can record it as such, with extra columns near the after-date indifference point.

3 Conjectures 2 and 3: the outside option and the value of a relationship

U and R enter the core model only through V_{end}, that is, through a and ψ. One lemma covers both.

Lemma 10 (Pivotality)**.** *Assume S1–S4. Beliefs about q and the budget are arbitrary. Let a parameter change from θ to θ' change a arbitrarily and raise ψ pointwise, $\psi_{\theta'} \geq \psi_\theta$ on $[0,1]$, leaving everything else unchanged. Let*

$$D_j = V_j^{\text{ask}} - V_{\text{end}}$$

be the value of asking minus the value of ending at step j; when $\hat{p}_j > 0$, $D_j = \hat{p}_j G_j$.

(i) If $\Psi_\theta(b) > 0$, so that the agent would ask for another date, then $D_j(b;\theta') \leq D_j(b;\theta)$.

(ii) If $\Psi_{\theta'}(b) \leq 0$, so that the agent would cut contact, then $D_j(b;\theta') \geq D_j(b;\theta)$.

Proof. By Lemma 3, the value of any policy π that asks now is

$$\Phi(\pi;\theta) = \tilde{\mathbb{E}}^\pi\Big[\sum \text{flows}\Big] + \tilde{\mathbb{E}}^\pi\big[V_{\text{end}}(b_\tau;\theta)\big].$$

Neither the flows nor $\tilde{\mathbb{P}}^\pi$ depend on θ; only the terminal payoff does. Let π and π' be ε-optimal asking policies at θ and θ', and write $\delta V(x) = V_{\text{end}}(x;\theta') - V_{\text{end}}(x;\theta)$. Then

$$\tilde{\mathbb{E}}^\pi[\delta V(b_\tau)] - \varepsilon \;\leq\; V_j^{\text{ask}}(\theta') - V_j^{\text{ask}}(\theta) \;\leq\; \tilde{\mathbb{E}}^{\pi'}[\delta V(b_\tau)] + \varepsilon.$$

By Lemma 2, $\delta V(x) = \mathbb{E}_x[a_{\theta'} - a_\theta] + \tilde{\kappa}\,\phi(x)$, where

$$\phi(x) = \Psi_{\theta'}(x)^+ - \Psi_\theta(x)^+, \qquad 0 \leq \phi(x) \leq \mathbb{E}_x\zeta, \qquad \zeta = \psi_{\theta'} - \psi_\theta \geq 0.$$

The first term of δV is linear in x, so by (2) it cancels against the same term in $\delta V(b)$.

(i) Here $\Psi_{\theta'}(b) \geq \Psi_\theta(b) > 0$, so $\phi(b) = \mathbb{E}_b\zeta$. Then

$$D_j(\theta') - D_j(\theta) \leq \tilde{\kappa}\big(\tilde{\mathbb{E}}^{\pi'}[\mathbb{E}_{b_\tau}\zeta] - \mathbb{E}_b\zeta\big) + \varepsilon = \varepsilon,$$

using (2) again.

(ii) Here $\Psi_\theta(b) \leq \Psi_{\theta'}(b) \leq 0$, so $\phi(b) = 0$. Then

$$D_j(\theta') - D_j(\theta) \geq \tilde{\kappa}\,\tilde{\mathbb{E}}^\pi[\phi(b_\tau)] - \varepsilon \geq -\varepsilon.$$

Let $\varepsilon \to 0$. □

The intuition is that the direct value of V_{end} is linear in beliefs, so it is the same, in expectation, whether the agent continues or ends. A parameter therefore changes the value of continuing only through the information value of the date for the after-date choice. That value rises when the parameter moves the after-date choice towards indifference, and falls when it moves the choice away.

Theorem 11 (Conjecture 2)**.** *Assume S1 and S3, and let $U < U'$.*

(a) Phase C (true). *The set of beliefs at which the agent asks for another date shrinks. The agent asks iff $\hat{p}(P_D) > 0$ and $V_D(b_{\text{yes}}) > U$, and*

$$V_D - U = \delta_D\mathbb{E}[hC]R - \big(1 - \delta_D(1 - \mathbb{E}h)\big)U$$

is strictly decreasing in U. This part does not need S4.

(b) On the day (true only where the agent would cut contact). *Assume S4 and let $A_j(U)$ be the set of beliefs at which the agent asks at step j.*

- *On $\{b : \Psi_U(b) \le 0\}$, where the agent would cut contact after the date: $A_j(U') \subseteq A_j(U)$. Raising U makes continuing harder, as conjectured.*
- *On $\{b : \Psi_{U'}(b) > 0\}$, where the agent would ask for another date even at U': $A_j(U) \subseteq A_j(U')$. Raising U makes continuing* easier*, contrary to the conjecture.*

(c) Spend. *While the budget does not bind, discretionary spend does not depend on U, by Proposition 1.*

Proof. (b) We have $\partial\psi/\partial U = -(1-h)\big(1-\delta_D(1-h)\big) \le 0$, so $\psi_U \ge \psi_{U'}$. Apply Lemma 10 with $\theta = U'$ and $\theta' = U$, recalling that the agent asks iff $\hat{p}_j > 0$ and $D_j \ge 0$, and that $\hat{p}_j$ does not depend on U. □

Example 12 (Conjecture 2 fails on the day, inside a threshold model)**.** Use the model of Proposition 8 with $R = 600$, $U = 100$ and a last step with $v = 100$ and $k = 32$. Corollary 7(b) holds, since $60 \ge 57.4$, so the asking set is a threshold.

- The threshold is $\pi^* = 0.1605$, where $\Psi = 4.24 > 0$.
- Raising U by 5 *lowers* the threshold to 0.1508.
- With $k = 25$ instead, the threshold is 0.0606, where $\Psi = -3.30 < 0$. There raising U by 5 raises it to 0.0622, as conjectured.

Theorem 13 (Conjecture 3, first part)**.** *Assume S1 and S3, and let $R < R'$.*

(a) Phase C (true). *The set of beliefs at which the agent asks for another date grows, because $V_D - U$ is increasing in R.*

(b) On the day. *Assume S4.*

- *On $\{b : \Psi_R(b) > 0\}$: $A_j(R') \subseteq A_j(R)$. A more valuable relationship makes continuing* harder*, contrary to the conjecture.*
- *On $\{b : \Psi_{R'}(b) \le 0\}$: $A_j(R) \subseteq A_j(R')$, as conjectured.*

In Example 12, raising R by 30 *raises the threshold from* 0.1605 *to* 0.1677*. With $k = 25$ it lowers the threshold from* 0.0606 *to* 0.0584*.*

Proof. $\partial\psi/\partial R = (1-h)\,\delta_D\,h\,c \ge 0$. Apply Lemma 10 with $\theta = R$ and $\theta' = R'$. □

Remark 14 (Why R does not simply favour escalation)*.* In the core model the chance of a relationship, $h(C, q_D)$, does not depend on how far up the ladder the date went. The direct value of a relationship, $\mathbb{E}[hC]R$, is therefore a martingale term that continuing cannot raise. If the intended reading is that escalating raises the chance of a relationship, h must depend on the step reached. Conjecture 3 would then need a separate argument.

Proposition 15 (Conjecture 3, second part: investment value)**.** *Assume S1–S4.*

(i) $I_j \ge 0$.

(ii) Two types*, with $x_0 = \psi(C_L) < 0 < x_1 = \psi(C_H)$: I_j is nondecreasing on $[0, \pi_k]$ and nonincreasing on $[\pi_k, 1]$, where $\pi_k = -x_0/(x_1 - x_0)$ is the after-date indifference point. Moreover $I_j(\pi) = 0$ whenever every posterior lies on the same side of π_k as π. If x_0, x_1 have the same sign, $I_j \equiv 0$.*

(iii) Settling: *for Beta beliefs, writing $\psi(c) = \sum_k \psi_k c^k$, which has degree at most* 3*,*

$$I_j\big(\mathrm{Beta}(\alpha, n-\alpha)\big) \le \tilde{\kappa}\,\frac{m_j}{n}\sum_{k\le 3} k|\psi_k| \;\longrightarrow\; 0 \qquad (n \to \infty).$$

(iv) "Largest when least settled" is false. *In the model of Proposition 8,* $I(1/2) = 0$ *but* $I(\pi_k) = 32.2$. *The consumption value depends on beliefs only through* $\mathbb{E}[C]$*, so the investment share of the step's value is zero at maximal uncertainty,* $\pi = 1/2$*. It is positive at* $\pi_k \approx 0.10$*, where the agent is fairly sure the match is incompatible. Nor is* I_j *monotone in precision at a fixed mean: with Beta beliefs of mean* 0.3*,* $I = 4.14$ *at* $n = 8$ *but* 4.72 *at* $n = 16$.

Proof. (ii) On $\{\Psi \le 0\}$, $I/\tilde{\kappa} = \sum_y (\cdot)^+$ with every summand nondecreasing, because its slope $P_y^H x_1 - P_y^L x_0$ is positive. On $\{\Psi \ge 0\}$, $I/\tilde{\kappa} = \sum_y (\cdot)^+ - \sum_y (\cdot) = \sum_y (\cdot)^-$, with every summand nonincreasing. If all posteriors lie on one side of π_k, $(\cdot)^+$ is linear on them and (2) gives $I = 0$.

(iii) Since $x \mapsto x^+$ is 1-Lipschitz, $I_j \le \tilde{\kappa}\, \mathbb{E}|\Psi(b') - \Psi(b)|$. For $b' = \mathrm{Beta}(\alpha + y, n - \alpha + m - y)$, each factor of $\mathbb{E}_{b'} C^k = \prod_{i<k} \frac{\alpha+y+i}{n+m+i}$ differs from the corresponding factor of $\mathbb{E}_b C^k$ by at most $m/(n+m+i) \le m/n$. All factors lie in $[0,1]$, so $|\mathbb{E}_{b'} C^k - \mathbb{E}_b C^k| \le km/n$.

(iv) From $\pi = 1/2$ the posteriors are 0.2 and 0.8, both above π_k, so (ii) gives $I = 0$. The other values are computed in `verify.py`. □

This is the classical point that the value of information is not monotone in prior uncertainty [1, 2]. The true statement is that the investment share is largest near the after-date indifference point, and that it vanishes as beliefs settle.

4 Conjecture 4: tight budgets

Proposition 16. *Assume S1 and S3.*

(i) Shadow value (true, by definition). *If* $V(b, \cdot)$ *is concave,* $\mu(B) = [V(B) - V(B - \varepsilon)]/\varepsilon$ *is nonincreasing in* B.

(ii) The premise usually fails. *Suppose step* j *has entry price* $k_j > 0$*,* $\hat{p}_j > 0$*, and* $G_j(b, \sigma k_j) > 0$*. Then* $V_j(b, \cdot)$ *jumps up at* $B = \sigma k_j$ *and is not concave on any interval containing* σk_j *in its interior.*

(iii) Thresholds (true in the core model, with no concavity needed). $G_j(b, \cdot)$ *is nondecreasing in* B *for every* b *and* j*. So the set of beliefs at which the agent asks shrinks as the budget falls.*

(iv) Spend (true under concavity). *If* $V_{j+1}(b', \cdot)$ *is concave for every* b'*, every selection of the optimal spend* $s_j^*(b, B)$ *can be taken nondecreasing in* B*. With less budget, the agent spends less, as if money were more expensive.*

(v) Cheaper opening activities (false). *The cost of the optimal opening activity need not rise with the budget, nor fall with the price of money. See Examples 17 and 18.*

(vi) Beyond the core. *Once the value after the date uses the budget, as in M2+ where another date must be paid for, thresholds can* fall *as the budget falls. See Example 20.*

Proof. (i) Slopes of chords of a concave function are nonincreasing.

(ii) For $B < \sigma k_j$ the step is unaffordable and $V_j = V_{\text{end}}$. At $B = \sigma k_j$, $V_j = V_{\text{end}} + \hat{p}_j G_j > V_{\text{end}}$. A nondecreasing function with an upward jump is not concave across it.

(iii) By induction, $V_{j+1}(b', \cdot)$ is nondecreasing: $V_{J+1} = V_{\text{end}}$ does not depend on B, and (1) preserves monotonicity. The feasible set $S_j(B)$ grows with B, and V_{end} does not depend on B.

(iv) For $s < s'$, $F(s', B) - F(s, B)$ consists of a part independent of B plus

$$\mathbb{E}\big[V_{j+1}(x) - V_{j+1}(x + d)\big], \qquad x = B - \sigma(k_j + s'), \quad d = \sigma(s' - s).$$

This is nondecreasing in B by concavity. So F has increasing differences in (s, B), $S_j(B) = [0, B/\sigma - k_j]$ is ascending, and Topkis' theorem applies. □

Example 17 (Opening activity and the budget)**.** Let C be known, $q = 1$, $\eta = 0$ and $\sigma = \lambda = 1$. Coffee has $x = \mathbb{E}[C]v = 10$ and $k = 8$; dinner has $x = 60$ and $k = 40$. Spend is optimal subject to the budget. A coffee opening leads to coffee then dinner; a dinner opening leads to dinner only.

- $B = 45$: coffee-first is worth 2.23, because dinner is no longer affordable after coffee. Dinner-first is worth 22.07. The agent opens with dinner.
- $B = 80$: coffee-first is worth 26.56 and dinner-first 24.33. The agent opens with coffee.

The richer agent chooses the cheaper opening. This $V(B)$ is not concave, as (ii) predicts.

Example 18 (Opening activity and the price of money)**.** Read "as if money were more expensive" as a rise in λ. Let coffee have $x = 10.8$ and $k = 8$, dinner $x = 63$ and $k = 40$, with $\sigma = 1$. Dinner is worth taking in either ladder for all λ near $\lambda_0 = 1.35$. Coffee-first minus dinner-first equals the coffee's own net value $W_c(\lambda)$ (Remark 4). This is 0.408 at $\lambda = 1.30$ and -0.4 at $\lambda = 1.40$. Dearer money switches the opening from coffee to dinner.

By the envelope theorem, $\partial V^a/\partial\lambda = -\sigma\,\mathbb{E}^a[\text{outlay}]$. A higher λ therefore favours the opening with the lower *expected total outlay*, which need not be the cheaper opening.

Remark 19. Under the conjecture's own premise, the overall value $V = \max(V^{\text{coffee}}, V^{\text{dinner}})$ is concave. Where the optimal opening switches transversally, a maximum of two differentiable functions has a convex kink. So the premise allows switches only at points of tangency, and cannot by itself deliver a systematic move to cheaper openings.

Example 20 (M2+: thresholds can fall with the budget)**.** At the last step the net value is $w = 10$ and the entry price $k = 20$, with $\sigma = 1$. Suppose the after-date option is worth 30 only if at least 40 of budget remains for another date. Then

$$G_J(B) = w + V_{\text{end}}(B - 20) - V_{\text{end}}(B).$$

This is 10 at $B = 30$, where a second date is unaffordable anyway; -20 at $B = 50$, where continuing would cost the second date; and 10 at $B = 70$. The agent asks at 30, ends at 50, and asks at 70.

5 Conjecture 5: spending to learn

Relax A7 as in Appendix A.7: the rating after step j comes from a family $\{P_m\}$ that is increasing in m in Blackwell's order, with $m = m(s)$ nondecreasing in spend. Integer Binomial m qualifies, as does Gaussian precision. Write

$$f_\mu(s) = \mu v_j g(s) - \sigma\lambda s.$$

Proposition 1's spend $s^0(\mu)$ is the unique maximiser of f_μ on $[0, \infty)$, which is strictly concave when $\mu v_j > 0$.

Proposition 21. *Assume S1–S4, a slack budget and $\hat{p}_j > 0$. Beliefs about q are arbitrary, and the sting model is any of D1, D2, KR.*

(i) Optimal spend is never below Proposition 1's spend. *Every optimal spend satisfies $s_j^*(b) \geq s^0(\mathbb{E}_b C)$, under all three sting models.*

(ii) It falls back as beliefs settle. *Suppose $m(\cdot)$ takes finitely many values, and beliefs settle: $b_n \to \delta_{\bar{c}}$ weakly, for example Beta beliefs with $n \to \infty$ and mean $\to \bar{c}$. Then $s_j^*(b_n) \to s^0(\bar{c})$.*

(iii) "Peaks under uncertainty" is false. *Take two types and the after-date model of Proposition 8, at the last step, with spend buying a binary signal of accuracy*

$$a(s) = \tfrac{1}{2} + 0.35(1 - e^{-s/20}) < 1 - \pi_k.$$

At $\pi = 1/2$, where uncertainty is greatest, the information premium is exactly zero: $s^* = s^0 = 0$. At $\pi_k = 0.104$, $s^* = 18.85$ while $s^0 = 0$.

Proof. (i) By Lemma 3, the optimal venue maximises $\Phi(s) = f_\mu(s) + e(s)$, with

$$e(s) = \mathbb{E}\big[V_{j+1}(b'_{m(s)}) \mid b\big].$$

V_{j+1} is convex in the belief about C (Theorem 24(a)), so Blackwell's theorem makes e nondecreasing. If $s < s^0$, then $f_\mu(s) < f_\mu(s^0)$ and $e(s) \le e(s^0)$, so s is not optimal.

(ii) We first show V_{j+1} is continuous in b for the weak topology on beliefs about C, including degenerate beliefs. Each ingredient is: $V_{\text{end}}(b) = \mathbb{E}_b a + \tilde{\kappa}\Psi(b)^+$; W_j, which depends on $\mathbb{E}_b C$; and the preposterior map $b \mapsto \sum_y P(y \mid b)V(b'_y)$. In the last of these, $P(y \mid b) = \mathbb{E}_b[P(y \mid C)]$ is continuous, the unnormalised posterior is weakly continuous, and $|P(y \mid b)V(b'_y)| \le P(y \mid b) \sup|V|$ near beliefs with $P(y \mid b) = 0$.

Hence $\text{EVSI}_m(b) = e_m(b) - V_{j+1}(b)$ is continuous, non-negative, and zero at degenerate beliefs. So

$$\varepsilon_n = \max_m \text{EVSI}_m(b_n) \to 0.$$

Maximisers of $\Phi_n = f_{\mu_n} + e$ lie in a fixed compact set, because $f_{\mu_n} \to -\infty$ linearly while e varies by at most ε_n. On that set, $f_{\mu_n} \to f_{\bar{c}}$ uniformly. Since $f_{\bar{c}}$ is strictly concave with unique maximiser $s^0(\bar{c})$, the standard argument applies: for $|s - s^0| \ge \epsilon$, $f_{\bar{c}}(s) \le f_{\bar{c}}(s^0) - \eta_\epsilon$. Once $2\varepsilon_n + 2\sup|f_{\mu_n} - f_{\bar{c}}| < \eta_\epsilon$, every maximiser of Φ_n is within ϵ of s^0.

(iii) From $\pi = 1/2$ the low posterior is $1 - a(s) > \pi_k$, so both posteriors lie above the kink and EVSI $\equiv 0$ for every s. At π_k, $\text{EVSI}(s) = \tilde{\kappa}(2a(s) - 1)\,\pi_k\,\psi(C_H)$ is strictly increasing in s. The optimal spends are computed in `verify.py`. □

Remark 22. The proposal's caveat that "with psychological costs of refusal EVSI is not guaranteed to be non-negative" does not bite here. The information that spend buys in A.7 is about compatibility, and Theorem 24(a) shows that such information is always weakly valuable in the core model, under all three sting models.

6 The value of information (Section 4.4)

Definition 23. Fix a belief b_C about C. A function V of beliefs about q is in the class $\mathcal{C}$ if it is nondecreasing in the likelihood-ratio order, and convex along every *MLR segment*, that is, along $t \mapsto (1-t)b_1 + tb_2$ for $b_1 \le_{\text{LR}} b_2$.

Theorem 24 (Sign of the value of information)**.** *Assume S1–S4.*

(a) Information about compatibility. *For every j, every B and every belief about q, V_j is convex in the belief about C. So every signal about C has non-negative EVSI under D1, D2 and KR. This includes ratings and the opening activity's signal in Definition 2.*

(b) Information about willingness. *Suppose the sting satisfies*

(K) *$f(p) = p - \kappa(p)$ is convex and twice differentiable, and $\kappa(p)/p$ is nonincreasing on $(0, 1]$.*

D1, D2 and KR all satisfy (K). Suppose also that the cues observed during the date are MLRP signals about q. Then every MLRP signal about q has non-negative EVSI at every decision point of the date.

(c) Non-monotone information about willingness can hurt. *Under D1, D2 and KR, some signals about q that are not MLR-ordered have strictly negative EVSI (Examples 25 and 26).*

Proof of (a). We argue by backward induction using Lemma 3, writing

$$V_j = \max\Big\{V_{\text{end}},\ \hat{p}_j^+\big(\sup_s[W_j(b;s) - c_j + \mathbb{E}(V_{j+1}(b') \mid b)]\big) + (1 - \hat{p}_j^+)V_{\text{end}}\Big\}.$$

Each ingredient is convex in b_C:

- V_{end} is convex (Lemma 2);
- $W_j(b; s)$ is affine in b_C;
- $b_C \mapsto \sum_y P(y \mid b_C) V_{j+1}(b'_y)$ is a sum of perspectives of the convex function V_{j+1}, hence convex;
- a supremum of convex functions is convex;
- $\hat{p}_j^+ \in [0, 1]$ does not depend on b_C, so the middle term is a convex combination of convex functions.

A maximum of convex functions is convex. Convexity of the value in beliefs is equivalent to non-negative EVSI for every signal. □

Proof of (b). Three facts about MLR segments are standard. Let $b_t = (1 - t)b_1 + tb_2$ with $b_1 \leq_{\text{LR}} b_2$.

- b_t is increasing in t in the likelihood-ratio order.
- For any likelihood $\ell(z \mid q)$, the posterior $b_{t,z}$ lies on the MLR segment from $b_{1,z}$ to $b_{2,z}$, with weight $tP(z \mid b_2)/P(z \mid b_t)$ on $b_{2,z}$.
- In particular, after a yes ($\ell = \varphi q$), the posterior lies on the segment from $b_{1,\text{yes}}$ to $b_{2,\text{yes}}$ at weight $s(t) = t\bar{p}_2/\bar{p}(t)$, where $\bar{p}(t) = \mathbb{E}_{b_t} q$ is affine and nondecreasing in t.

Step 1: expectations over MLRP signals preserve $\mathcal{C}$. Let $V \in \mathcal{C}$ and let z be an MLRP signal. Along a segment,

$$\mathbb{E}[V(b_{t,z})] = \sum_z \big(A_z(t) + B_z(t)\big)\, \phi_z\Big(\frac{B_z(t)}{A_z(t) + B_z(t)}\Big),$$

with $A_z(t) = (1 - t)P(z \mid b_1)$ and $B_z(t) = tP(z \mid b_2)$ affine, and ϕ_z convex. This is a perspective, hence convex in t. Monotonicity follows from MLRP. Expectations over ratings (independent of q) and suprema over spend also preserve $\mathcal{C}$.

Step 2: the ask step preserves $\mathcal{C}$. Suppose $X \in \mathcal{C}$, and recall from Lemma 3 that $V_j - V_{\text{end}} = f(p)^+ X(b_{\text{yes}})^+$.

Monotonicity: by (K), $f = p \cdot (f/p)$ is nondecreasing where positive; p is nondecreasing in the order; and $b \mapsto b_{\text{yes}}$ is monotone.

Convexity along a segment: let $g(t) = f(p(t))\, \phi(s(t))$, where $\phi(s) = X$ along the yes-segment is convex and nondecreasing. Assume $\phi \in C^2$ for now. Since $s' = \bar{p}_1\bar{p}_2/\bar{p}^2 \geq 0$ and $s'' = -2\bar{p}'s'/\bar{p}$,

$$g'' = f''\, p'^2\, \phi + f\, \phi''\, s'^2 + 2\, p'\phi' s' \Big(f' - \frac{f}{p}\Big).$$

On the set where $f > 0$ and $\phi > 0$, every term is non-negative:

- $f'' \geq 0$ by (K);
- $p' \geq 0$, because LR order implies ordered means;
- $\phi' \geq 0$ and $\phi'' \geq 0$;
- $f' - f/p = p\,(f/p)' \geq 0$ by (K).

A general convex nondecreasing ϕ is a uniform limit of C^2 ones, and convexity survives limits. The set where $f > 0$ and $\phi > 0$ is an interval $[t_0, 1]$, and g vanishes at t_0 if $t_0 > 0$. So $f^+\phi^+$ is 0 on $[0, t_0]$ and convex nondecreasing on $[t_0, 1]$, hence convex on $[0, 1]$.

Induction on j, starting from $V_{J+1} - V_{\text{end}} \equiv 0$, gives $V_j(b_C, \cdot) - V_{\text{end}}(b_C) \in \mathcal{C}$.

Step 3: from convexity on segments to non-negative EVSI. Let $z \in \{z_1 < \cdots < z_K\}$ be an MLRP signal. Reveal it in stages: first whether $z = z_K$, then, given $z < z_K$, whether $z = z_{K-1}$, and so on. At each stage MLRP implies $b_{\{z=z_k\}} \geq_{\text{LR}} b_{\{z<z_k\}}$, and the current belief lies on the MLR segment between them. So the expected value weakly rises at each stage. The tower property gives $\mathbb{E}[V(b_z)] \geq V(b)$. □

Example 25 (Negative EVSI under D1)**.** Let C be known, normalise $V_{\text{end}} = 0$, and take two steps left with net values $w_1 = 0$ and $w_2 = 1$. Let $\varphi_2 = 1$, D1 with $\eta = 1/2$, so $f(p) = 1.5p - 0.5$.

Let $q \in \{0.2, 0.6, 1\}$ with prior $(0.3, 0.5, 0.2)$. The signal reveals whether q is extreme, $\{0.2, 1\}$, or middling, $\{0.6\}$. It is not MLR-ordered. The value is

$$V(b) = f(\mathbb{E}_b q)\big(w_1 + f(\mathbb{E}_b[q \mid \text{yes}])\, w_2\big).$$

- Prior: $\mathbb{E}q = 0.56$ and $\mathbb{E}[q \mid \text{yes}] = 0.7$, so $V = 0.34 \times 0.55 = 0.1870$.
- Posterior "extreme", probability $1/2$, $b_1 = (0.6, 0, 0.4)$: $V = 0.28 \times 0.7231 = 0.2025$.
- Posterior "middling", $b_2 = \delta_{0.6}$: $V = 0.4 \times 0.4 = 0.16$.

So EVSI $= 0.1812 - 0.1870 = -0.0058 < 0$. With $\eta = 0$ the EVSI is exactly zero.

The signal moves the chance of a yes now, $\mathbb{E}q$, and the chance of a yes later, $\mathbb{E}[q \mid \text{yes}]$, in opposite directions. That breaks the third term of g'' in the proof of (b).

Example 26 (Negative EVSI under D2 and KR)**.** Let $q \in \{0.1, 0.8, 1\}$ with prior $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$, and use the same "extreme versus middling" signal. Take D2 with $\eta = 2$, equivalently KR with $\eta(\Lambda - 1) = 2$, so $f(p) = p(2p - 1)$; and take $w_1 = -\frac{1}{2}$, $w_2 = 1$, $\varphi_2 = 1$.

- Prior: $V = 0.2363 \times 0.0906 = 0.0214$.
- Posterior "extreme": $V = 0.055 \times 0.2679 = 0.0147$.
- Posterior "middling": $V = 0$, because step 2 is no longer worth asking for and step 1 is not worth taking alone.

So EVSI $= 0.0074 - 0.0214 = -0.0140 < 0$.

Answer to the open question. In the core model, the value of information can be negative only for information about willingness that is not ordered by the likelihood ratio. Ratings, and cues of the kind used in Section 5.4, which are MLRP, always have non-negative value. This is consistent with the proposal's numerical search, which found no counterexample.

7 The conjectural prediction about opening activities

Prediction 6 of Appendix A.9 says that more uncertainty before the date favours a more informative opening activity, depending on where beliefs sit relative to the decision threshold. As an unconditional statement this is false, for the same reason as Proposition 15(iv): at maximal uncertainty the EVSI of both openings can be zero. What is true follows from Theorem 24(a), Blackwell's theorem and Proposition 15:

- the informativeness premium of a Blackwell-more-informative opening is never negative;
- it vanishes as beliefs settle;
- with two types it is largest near the relevant indifference point.

8 Suggested changes to the proposal

1. Replace Conjecture 1 by Theorem 6 with Corollary 7, and cite Proposition 8. Where (M) fails, the card may need more than one boundary near the after-date indifference point.

2. Replace Conjectures 2 and 3 (first part) by Lemma 10 and Theorems 11–13. Keep the Phase C statements, which are true. Note that in the core model R affects escalation only through information, because h does not depend on the step reached. Prediction 2 (market thickness) and Prediction 5 (relationship value) of Appendix A.9 should be restated accordingly.

3. Replace "largest when beliefs are least settled", in Conjectures 3 and 5 and in Prediction 6, by "largest near the after-date indifference point, and vanishing as beliefs settle".

4. In Conjecture 4, drop the concavity premise, which fails whenever entry prices bind. State Proposition 16(iii)–(iv) instead, and drop "cheaper opening activities" from the conjecture and from Prediction 3.

5. In Section 4.4 and the "Convexity" bullet of Section 8, replace the open question with Theorem 24.

6. Technical points:
   - state the convention $\kappa\Delta^{+}$;
   - note that the fractional likelihood is a quasi-likelihood, so results relying on the martingale property are exact only with integer m_a;
   - record the closed form in Remark 4;
   - record that $\ell = 0$ throughout the core model (Lemma 3), so the decision to ask on the day reduces to $p \geq p^{*}_{m}(0)$ together with $G \geq 0$.

Numerical verification

`python3 verify.py` reproduces every number above and runs randomised checks of:

- Corollary 7(b)–(c): monotone G_j whenever the condition holds;
- the sign pattern of Lemma 10 for U and R, at several thousand belief points;
- Proposition 16(iii);
- Theorem 24(a), as convexity of V_0 in beliefs about C;
- Theorem 24(b), as non-negative EVSI of Binomial cues about q under general priors.

The output is saved in `verify_output.txt`.

References

[1] Gould, J. P. (1974). Risk, stochastic preference, and the value of information. *Journal of Economic Theory* 8(1), 64–84.

[2] Hilton, R. W. (1981). The determinants of information value: synthesizing some general results. *Management Science* 27(1), 57–64.

[3] Topkis, D. M. (1978). Minimizing a submodular function on a lattice. *Operations Research* 26(2), 305–321.

[4] Milgrom, P. R. (1981). Good news and bad news: representation theorems and applications. *Bell Journal of Economics* 12(2), 380–391.

[5] Blackwell, D. (1953). Equivalent comparisons of experiments. *Annals of Mathematical Statistics* 24(2), 265–272.